Computer Networks Lab 07

Course: Computer Networks (CL3001) **Semester:** Fall 2026
Instructor: Sadaf Zehra **T.A:** N/A

Note:

- Maintain discipline during the lab.
- Listen and follow the instructions as they are given.
- Just raise hand if you have any problem.
- Completing all tasks of each lab is compulsory.
- Get your lab checked at the end of the session.

Lab Objective

Computer Networks CL3001	LAB - 07 ACL and Wireshark

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES,
KARACHI CAMPUS
FAST SCHOOL OF COMPUTING (AI & DS, CS, CY, SE) Fall 2026

- Introduction to ACL & configuration of ACL in Cisco Packet Tracer.

Note:

Submit the screenshot of every step and also submit a .pkt file.

Lab Exercise – I

Implement the topology given below on cisco packet tracer:

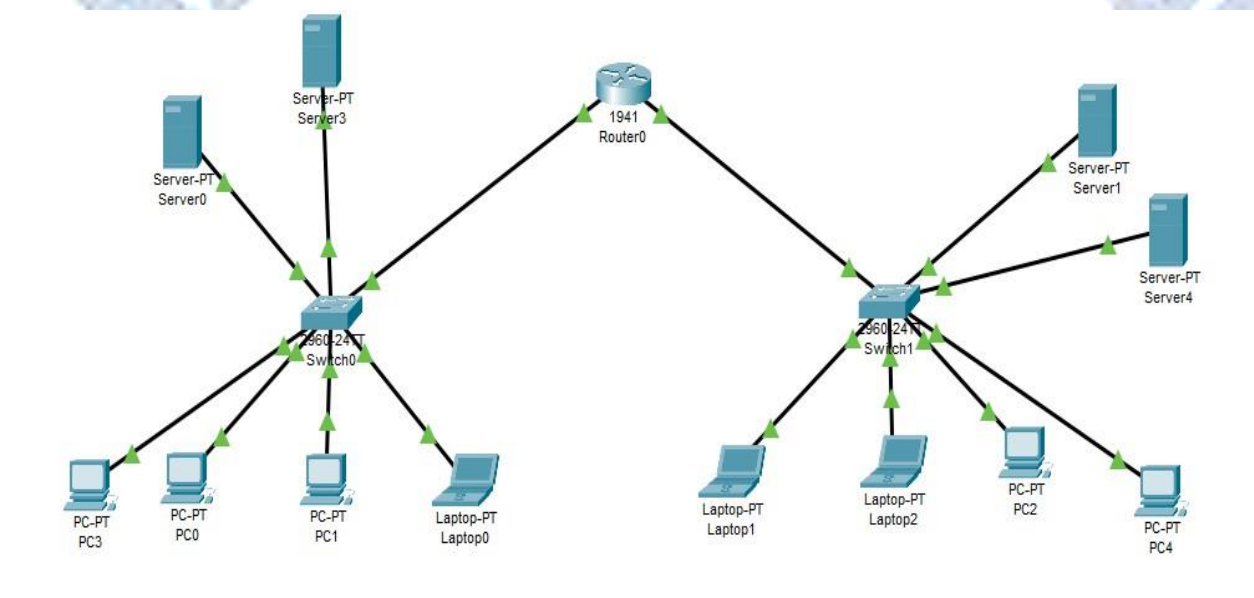


Figure 01

Tasks:

- Enhance the topology by adding one router in the network.
- Prevent PCs in the LEFT NETWORK from communicating with any device in the RIGHT NETWORK. However, the RIGHT NETWORK should be able to communicate with the LEFT NETWORK.
- Allow only THREE PCs in the LEFT NETWORK to access and communicate with the RIGHT NETWORK, while blocking all other devices in the LEFT NETWORK.
- Allow PCs in the RIGHT NETWORK to communicate with the LEFT NETWORK, but block communication between two specific PCs in the LEFT NETWORK.
- Allow only TWO PCs in the RIGHT NETWORK to communicate with the LEFT NETWORK, while preventing all other PCs in the RIGHT NETWORK from accessing the LEFT NETWORK.

Lab Exercise – II

Implement the topology given below on cisco packet tracer:

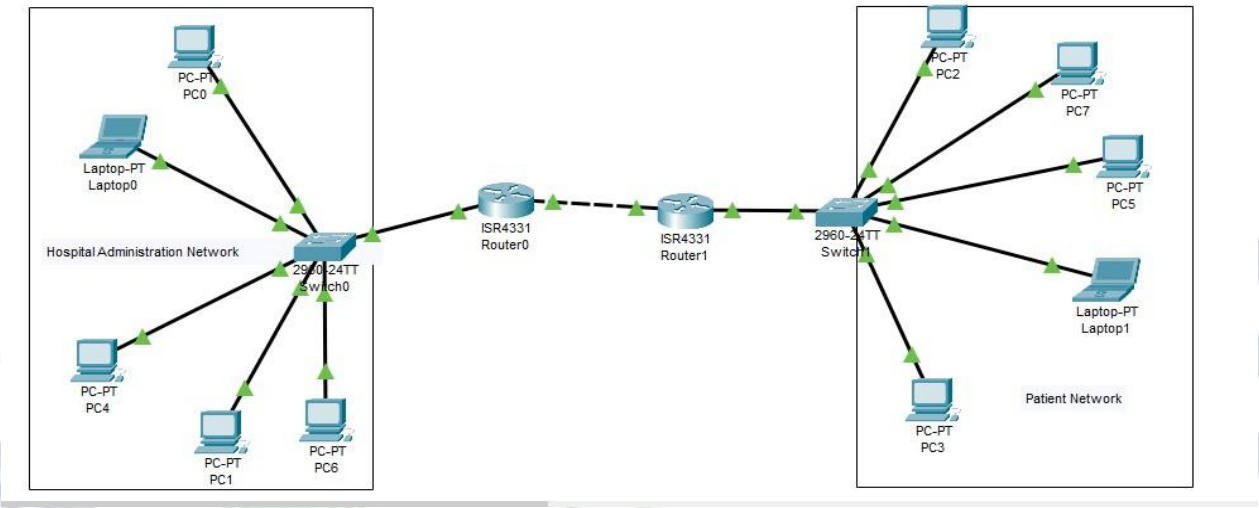


Figure 02

Tasks:

- Prevent PCs in the PATIENT network from communicating with any device in the HOSPITAL ADMINISTRATION network. However, HOSPITAL ADMINISTRATION should be able to communicate with the PATIENT network.
- Allow only THREE PCs in the PATIENT network to access and communicate with the HOSPITAL ADMINISTRATION network, blocking every other device in the PATIENT network.
- Prevent PCs in the HOSPITAL ADMINISTRATION network from communicating with any device in the PATIENT network. However, the PATIENT network should be able to communicate with the HOSPITAL ADMINISTRATION network.
- Allow only TWO PCs in the HOSPITAL ADMINISTRATION network to access and communicate with the PATIENT network, blocking every other device in the HOSPITAL ADMINISTRATION network.
- Block communication between one specific PC in the PATIENT network and all devices in the HOSPITAL ADMINISTRATION network, while allowing all other PCs in the PATIENT network to communicate with the HOSPITAL ADMINISTRATION network.
- Allow PCs in the PATIENT network to communicate with the HOSPITAL ADMINISTRATION network, but block access from the HOSPITAL ADMINISTRATION network to one specific PC in the PATIENT network.

Lab Exercise Wireshark:

Open the WiresharkLab5H.pcap file on Wireshark. Apply a UDP filter and select the packet having serial no. 15 and answer the following questions:

1. What is the domain name being queried in the packet?

2. What is the destination and source port no?

3. What is the destination and source ip address of the packet?

4. Determine how many fields are there in the UDP header?

5. What is the maximum number of bytes that can be included in a UDP payload?